

# Employee Data Management using ServiceNow Import Sets and Transform Maps

## Project Report

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Team ID: SWTID-2026-6228

| Name                | Role      |
|---------------------|-----------|
| Sabibaa Bhanu B     | Team Lead |
| Sridevi G           | Member    |
| Kotteeswari G       | Member    |
| Aswathi S           | Member    |
| Potturu Charuhasini | Member    |

# 1. INTRODUCTION

## 1.1 Project Overview

This project implements an automated employee data management solution on the ServiceNow platform. Employee data prepared in a spreadsheet (.xlsx) is uploaded into a temporary Import Set Table, validated, and moved into a custom target table, Employee Test (u\_employee\_test), using a Transform Map. Coalesce on Employee ID prevents duplicate records, so repeated imports update existing employees instead of inserting copies. Reports and a dashboard give visibility into the resulting data.

## 1.2 Purpose

- Replace manual employee data entry with a repeatable bulk import process.
- Maintain accurate, duplicate-free employee records across repeated imports.
- Give administrators and HR teams a single dashboard for employee data insights.

# 2. IDEATION PHASE

## 2.1 Problem Statement

In real projects the same employee data is received repeatedly from external sources. Entering it manually is slow and error-prone, and re-importing it can create duplicate or inconsistent records. A reliable, validated, duplicate-safe import process is needed, along with reporting on the imported data.

## 2.2 Empathy Map Canvas

| Quadrant | Description                                                     |
|----------|-----------------------------------------------------------------|
| Says     | "I need employee records to be accurate and up to date."        |
| Thinks   | Manual uploads risk duplicates and missing fields.              |
| Does     | Maintains spreadsheets, uploads files, corrects errors by hand. |
| Feels    | Frustrated by rework; wants confidence in data quality.         |

## 2.3 Brainstorming

- Use a custom table (Employee Test) to hold employee records.
- Use Import Sets as a staging area before data reaches the target table.
- Use a Transform Map with Employee ID as the Coalesce field.
- Build reports (by Department, by Location, full list) and combine them into a dashboard.

# 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

1. Prepare: create the spreadsheet with employee data.
2. Upload: load the file via System Import Sets > Load Data.
3. Transform: run the Transform Map to insert or update records.
4. Verify: check the Transform History and the Employee Test table.
5. Analyse: view reports on the Employee Analytics Dashboard.

### 3.2 Solution Requirement

#### Functional requirements

- Custom table with fields: Employee ID, Employee Name, Email, Department, Location.
- Import Set Table for temporary staging of uploaded data.
- Transform Map with field mapping and Coalesce on Employee ID.
- Validation of required fields (Employee ID, Name, Email).
- Reports and a dashboard on the Employee Test table.

**Non-functional requirements:** data integrity, no duplicate records, repeatable imports, role-based sharing of reports.

### 3.3 Data Flow Diagram

| Source               | Load Data   | Import Set Table  | Transform Map + Coalesce                  | Employee Test table    | Reports and Dashboard         |
|----------------------|-------------|-------------------|-------------------------------------------|------------------------|-------------------------------|
| Google Sheet / Excel | Upload file | Temporary staging | Validate; insert or update by Employee ID | Final employee records | Employee Analytics Dashboards |

### 3.4 Technology Stack

| Component    | Technology                                      |
|--------------|-------------------------------------------------|
| Platform     | ServiceNow                                      |
| Source data  | Google Sheets, exported as Excel (.xlsx)        |
| Data loading | System Import Sets: Load Data, Import Set Table |
| Data mapping | Transform Map with Field Maps and Coalesce      |
| Target table | Employee Test (u_employee_test)                 |
| Reporting    | ServiceNow Reports and Dashboards               |

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The problem is duplicate-prone, manual employee data handling. Import Sets with Coalesce solve it directly: data is staged, validated, and then inserted or updated based on Employee ID, keeping the database clean even when the same file is imported repeatedly.

### 4.2 Proposed Solution

6. Spreadsheet: create a Google Sheet with sample data and download it as Sample Spreadsheet.xlsx.
7. Custom table: create Employee Test (u\_employee\_test) via Tables > Create New, and add String fields Employee ID, Employee Name, Email, Department and Location using Form Layout.
8. Import Set Table: use Load Data to create and load the staging table (Employee Import) from the spreadsheet (sheet 1, header row 1).
9. Transform Map: name it Sample Spreadsheet Import; source Employee Import; target Employee Test; use Auto Map Matching Fields and save.
10. Transform and validate: run the transform, confirm the success message, and check the records in Employee Test.
11. Coalesce: set Coalesce to true on the Employee ID field map to avoid duplicates.
12. Reports and dashboard: create three reports and add them to the Employee Analytics Dashboards.

### 4.3 Solution Architecture

The architecture has four layers: Source (spreadsheet), Staging (Import Set Table), Processing (Transform Map with Coalesce and validation), and Target and Presentation (Employee Test table, reports, dashboard). Employee ID is the unique key that decides whether a row is inserted or updated.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Milestone                         | Activities                                                          | Status |
|-----------------------------------|---------------------------------------------------------------------|--------|
| 1. Spreadsheet and table creation | Create sheet and sample data; create Employee Test table and fields | Done   |
| 2. Import Set and Transform Map   | Load data; create Transform Map; auto map fields                    | Done   |
| 3. Transform, validate, coalesce  | Run transform; verify records; enable Coalesce on Employee ID       | Done   |
| 4. Reports and dashboard          | Create 3 reports; build Employee Analytics Dashboards               | Done   |

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

| Test case                   | Input                                                     | Expected result                      | Actual result                          |
|-----------------------------|-----------------------------------------------------------|--------------------------------------|----------------------------------------|
| Initial import              | Sample spreadsheet                                        | All rows inserted into Employee Test | Pass: all records created              |
| Re-import with changes      | 4 rows: 2 existing IDs with changed Name/Email, 2 new IDs | 2 inserted, 2 updated                | Pass: 2 inserted, 2 updated            |
| Re-import of identical file | Same 4 rows                                               | No inserts or updates                | Pass: 0 inserted, 0 updated, 4 ignored |

The tests confirm that Coalesce on Employee ID updates changed records, inserts only genuinely new ones, and ignores unchanged data.

## 7. RESULTS

### 7.1 Output Screenshots

All records were successfully imported into ServiceNow using the Import Set and Transform Map. The six employee records were created in the Employee Test table, and the dashboard reports reflect the imported data.

The screenshot displays the ServiceNow 'Load Data' configuration page. The browser address bar shows the URL: dev443751.service-now.com/now/nav/ui/classic/params/target/create\_import\_set.do. The page header includes the ServiceNow logo and navigation links: All, Favorites, History, Workspaces, and a 'Load Data' button. Below the header, the 'Import set table' section has two radio buttons: 'Create table' (selected) and 'Existing table'. A red asterisk indicates a required field for 'Label', which is set to 'Employee Import'. The 'Name' field is set to 'u\_employee\_import'. The 'Source of the import' section has two radio buttons: 'File' (selected) and 'Data source'. Under 'File', the 'Choose File' button is next to the text 'sample sheet 23.xlsx'. The 'Sheet number' field is set to '1' and the 'Header row' field is set to '1'. A 'Submit' button is located at the bottom of the form. The Windows taskbar at the bottom shows the system clock as 9:34 PM on 10/3/2025.

Figure 1: Load Data - creating the Employee Import (u\_employee\_import) table from the spreadsheet

dev443751.service-now.com/how/nav/ui/classic/params/target/create\_import\_set.do

**servicenow** All Favorites History Workspaces Load Data Search

Load Data Submit

Import set table ☐ Create table ☒ Existing table

Import set table Employee Import [u\_employee\_import]

Source of the import ☒ File ☐ Data source

File Choose File sample sheet 23.xlsx

Sheet number 1

Header row 1

Submit

Figure 2: Load Data - loading new spreadsheet data into the existing Employee Import table

**servicenow** All Favorites History Admin Employee Tests Search Actions on selected rows... New

Employee Tests Employee ID Search

All

| Employee ID | Employee Name | Department | Email            | Location  |
|-------------|---------------|------------|------------------|-----------|
| SB0007      | Sushanth      | Servicenow | test07@gmail.com | Tamilnadu |
| SB0003      | Ajay          | Servicenow | test03@gmail.com | Tamilnadu |
| SB0010      | Abhishek      | Servicenow | test10@gmail.com | Tamilnadu |
| SB0006      | Prashanth     | Servicenow | test06@gmail.com | Tamilnadu |
| SB0002      | Sandeep       | Servicenow | test02@gmail.com | Tamilnadu |
| SB0005      | Yashwanth     | Servicenow | test05@gmail.com | Tamilnadu |
| SB0009      | Praneeth      | Servicenow | test09@gmail.com | Tamilnadu |
| SB0001      | Ratan Kumar   | Servicenow | test01@gmail.com | Tamilnadu |
| SB0008      | Ravi          | Servicenow | test08@gmail.com | Tamilnadu |
| SB0004      | Arya          | Servicenow | test04@gmail.com | Tamilnadu |
| SB0011      | Arnav         | Salesforce | test22@gmail.com | Tamilnadu |

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Figure 3: Employee Test table showing the imported records

| Name                                  | Decision Type | Operation   | Type   | Active | Updated by          | Updated             |
|---------------------------------------|---------------|-------------|--------|--------|---------------------|---------------------|
| pa_dashboards                         | Allow If      | write       | record | true   | admin               | 2019-03-25 09:32:03 |
| pa_dashboards                         | Allow If      | write       | record | true   | admin               | 2016-04-19 03:29:14 |
| pa_dashboards                         | Allow If      | read        | record | true   | admin               | 2019-03-25 09:34:18 |
| pa_dashboards                         | Allow If      | report_view | record | true   | admin               | 2022-11-16 22:19:19 |
| pa_dashboards                         | Allow If      | delete      | record | true   | admin               | 2016-04-19 03:26:55 |
| pa_dashboards                         | Allow If      | read        | record | true   | admin               | 2018-03-01 07:27:17 |
| pa_dashboards                         | Allow If      | create      | record | true   | admin               | 2026-10-02 04:15:38 |
| pa_dashboards                         | Allow If      | delete      | record | true   | admin               | 2019-03-25 09:35:29 |
| pa_dashboards.*                       | Allow If      | query_range | record | true   | @@snc_write_audit@@ | 2026-06-13 04:38:32 |
| pa_dashboards.group                   | Allow If      | write       | record | true   | admin               | 2016-07-28 06:29:53 |
| pa_dashboards.notification_link_label | Allow If      | query_range | record | true   | @@snc_write_audit@@ | 2026-06-13 04:38:32 |
| pa_dashboards.notification_text       | Allow If      | query_range | record | true   | @@snc_write_audit@@ | 2026-06-13 04:38:32 |
| pa_dashboards.owner                   | Allow If      | write       | record | true   | admin               | 2021-11-04 08:32:01 |

Figure 4: Access Controls (ACL) for pa\_dashboards used to enable dashboard creation

| Employee ID             | Employee Name | Department | Email | Location |
|-------------------------|---------------|------------|-------|----------|
| Employee ID: SB0001 (1) |               |            |       |          |
| Employee ID: SB0002 (1) |               |            |       |          |
| Employee ID: SB0003 (1) |               |            |       |          |
| Employee ID: SB0004 (1) |               |            |       |          |
| Employee ID: SB0005 (1) |               |            |       |          |

Employees by Location

Count: 6

Figure 5: Employee Analytics Dashboard - Employee List Report

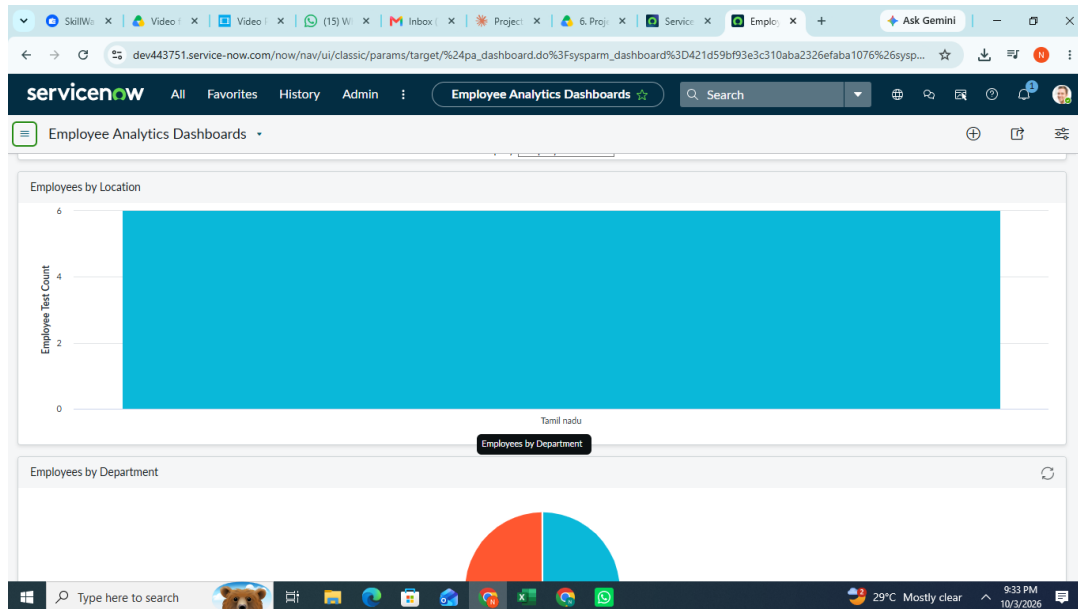


Figure 6: Employee Analytics Dashboard - Employees by Location

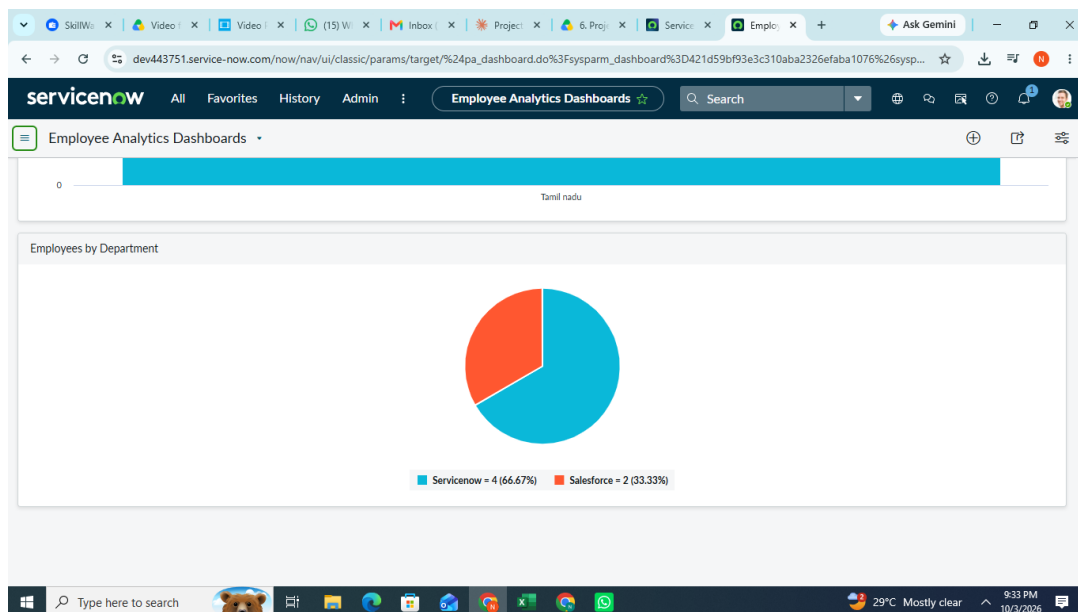


Figure 7: Employee Analytics Dashboard - Employees by Department (Servicenow 4, Salesforce 2)

## 8. ADVANTAGES & DISADVANTAGES

| Advantages                                  | Disadvantages                                   |
|---------------------------------------------|-------------------------------------------------|
| Prevents duplicate records through Coalesce | Spreadsheet column names must match the mapping |
| Fast bulk loading instead of manual entry   | Coalesce depends on a clean, unique Employee ID |

| Advantages                                          | Disadvantages                                                |
|-----------------------------------------------------|--------------------------------------------------------------|
| Staging table allows validation before final insert | Needs admin access (e.g., ACL change for dashboard creation) |
| Dashboard gives real-time visibility                | Manual upload step is still required each time               |
| Reports can be restricted by user, group or role    |                                                              |

## 9. CONCLUSION

This project delivered an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency and data integrity during employee onboarding and updates. Enabling Coalesce on Employee ID means every import identifies existing employees and updates them rather than inserting redundant entries, which keeps the database clean and consistent.

Reports and a dashboard were added for insight into employee data: department-wise distribution, location-wise distribution and the full employee list. Consolidated on one dashboard, they help administrators and HR teams monitor data trends and quality and reduce manual analysis.

## 10. FUTURE SCOPE

- Schedule automatic imports (scheduled data imports) from a shared source.
- Add data-quality validation using Transform Map scripts.
- Add reports for import success/error counts, newly added and updated employees.
- Send notifications to HR when an import fails.
- Integrate with HR systems through REST APIs or Integration Hub.

## 11. APPENDIX

**Source Code:** Not applicable (configuration-based, no custom code).

**Dataset Link:** [Sample Spreadsheet - Google Sheets](#)

**GitHub Link:** [Sabibaa-Bhanu/Import-Data-using-Transform-Maps-Spreadsheet-: Import Data using Transform Maps \(Spreadsheet\)](#)

**Project Demo Link:**